

1. ABSTRACT

An Event and Room Occupancy Management System for managing campus events, room allocations, registrations and occupancy. The system uses a React frontend, API Gateway, Node.js/Express microservices and separate MySQL data bases for user, admin and event data.

Outcome:

Centralized management and room conflict prevention, registration tracking and occupancy monitoring.

2. PROBLEM STATEMENT & OBJECTIVES

Problem: Campus event and room management can become difficult when event schedules, room availability, registrations and occupancy data are handled separately.

Objectives:

- Manage events and rooms
- Prevent invalid/conflicting room assignments
- Track student registrations and occupancy
- Provide secure room-based access through REST APIs

Scope: Student and admin workflows covering events, rooms, bookings, registrations, availability and occupancy.

3. EXISTING VS PROPOSED SYSTEM

Existing (Manual)	Proposed (System)
Manual/separate tracking	Manual/separate web application
Single backend/ database	API Gateway + service-based architecture with separate databases
Manual room scheduling	Availability and conflict/conflict validation
Static occupancy values	Registration-aware effective occupancy monitoring

4. FEASIBILITY STUDY

Technical: React, Node.js, Express, MySQL, REST APIs, JWT, bcrypt, Axios and VS Code.

Economic: Open-source technologies and Local development keep implementation cost low.

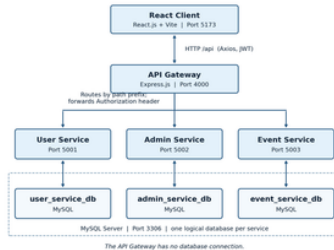
Operational: Role-based web workflows simplify event, room and registration management.

Schedule: Designed and implemented with the semester PBL timeline.

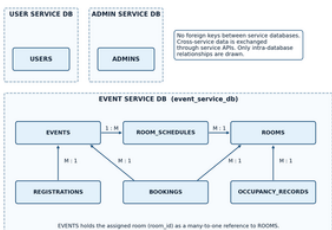
5. TOOLS & TECHNOLOGIES

	Front End React.js, Vite, HTML, CSS,JS
	Back End Node.js, Express, JS
	Data Base MySQL 8.0, separate database
	Security JWT authentication, bcrypt password hashing
	Tools VS,Code, Postman, Git/GitHub, MySQL, Workbench

6. SYSTEM ARCHITECTURE

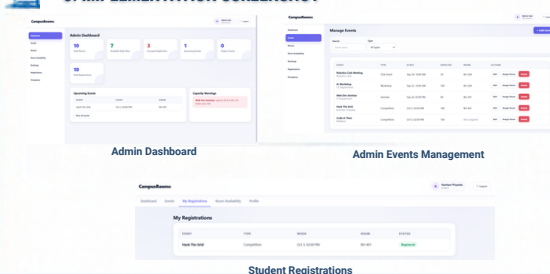


7. ER DIAGRAM & DATABASE DESIGN



Service databases are logically separated, the event service manages event, room, schedule, booking, registration and occupancy data. No cross-service foreign keys are used.

8. IMPLEMENTATION SCREENSHOT



9. RESULTS & TESTING

Core CRUD workflows, authentication, room assignment, schedule/conflict validation, registrations, room availability and occupancy monitoring were tested. Room assignment rejects inactive rooms, Insufficient capacity and overlapping schedules. Occupancy updates reflect current registered participants.

10. CONCLUSION

The system provides a structured web platform for campus event and room management, combining role-based access, service separation, MySQL data management, registration-aware occupancy monitoring.

11. FUTURE SCOPE

- Email/SMS notifications
- Calendar integration
- Live sensor-based occupancy
- Cloud deployment and centralized monitoring
- Advanced analytics and reports

Github Repository QR



12. REFERENCES

- React – <https://react.dev/>
- Vite – <https://vite.dev/>
- Node.js – <https://nodejs.org/>
- Express.js – <https://expressjs.com/>
- MySQL 8.0 – <https://dev.mysql.com/doc/mysql/8.0/en/>
- Axios – <https://axios-http.com/>
- JWT – <https://jwt.io/>
- bcrypt – <https://www.npmjs.com/package/bcrypt>
- Postman – <https://www.postman.com/>
- MDN – <https://developer.mozilla.org/>

